

Should We Automate Incident Routing?

A business case built on 24,918 real IT incidents

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Decision requested. Do **not** fund an automated routing project: on this data, no router beats today's routing (56% for a rules table and 61% for machine learning, against 66% today). Instead, approve a four-week pilot of **one assignment rule for Category 23**, plus a root-cause check first. On later incidents the rule reached the right team 42% of the time, against 7% today. The change is a configuration edit, estimated at about 8 hours.

41%

of incidents change teams at least once

+62.5 h

median extra time per hand-off, like for like

66% vs 61%

today's routing vs the best router tested

84.5%

of Category 23 tickets change teams

1. The business problem

When an IT ticket is opened, it is assigned to a support group. If that group cannot resolve it, the ticket is passed to another, and sometimes to several. Every hand-off is a delay for the user waiting and work for the analysts reading and re-routing the ticket. The question for the IT Service Desk Manager is whether to invest in automating the routing, and if so where.

2. The data

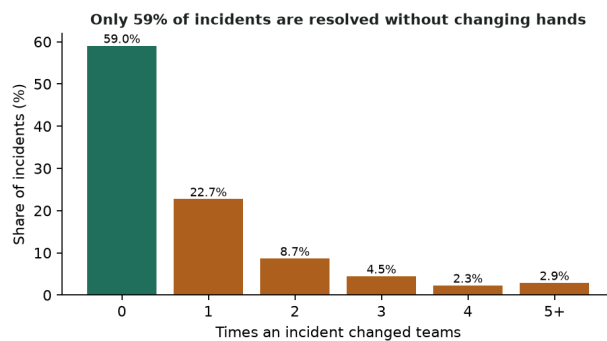
A real, anonymized ServiceNow log from an IT company (Amaral, Fantinato and Peres, 2018; UCI Machine Learning Repository; CC BY 4.0): **24,918 incidents** and 141,712 logged updates. 98% of incidents were opened in three months (March to May 2016), about 8,146 a month, so figures are reported per month rather than scaled up to a year. 23,362 incidents have a resolution time. Group, category and agent names are anonymized (Group 20, Category 23).

3. What the data shows

3.1 Hand-offs are common and costly

41% of incidents changed teams at least once and 18% changed twice or more, adding up to 19,084 hand-offs. Incidents that changed hands took a median 82.5 hours against 1.2, and met their SLA 46% of the time against 73%.

That raw gap overstates the cost, because harder tickets are passed around more. Comparing incidents that ended up in the same resolving group (35 groups, covering 93% of hand-off incidents), a hand-off is associated with **+62.5 median hours** and **17.3 fewer SLA points**. This is an association, not proof that the hand-off caused it.



Share of incidents by number of hand-offs.



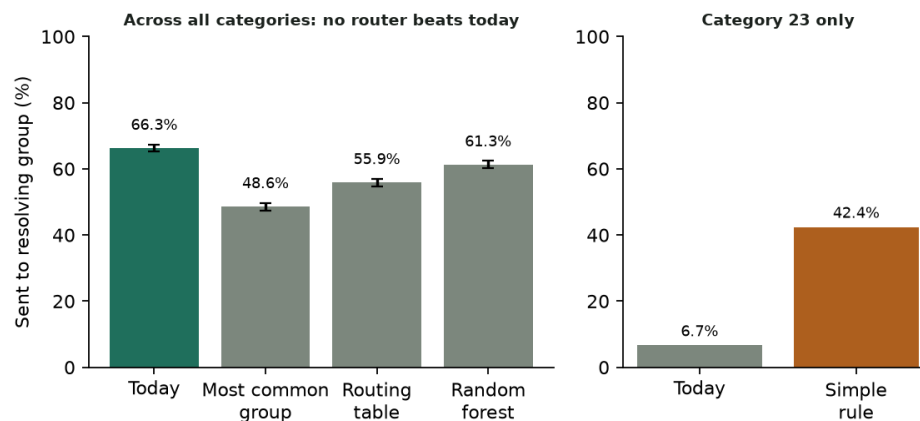
Raw comparison; the like-for-like gap is smaller.

3.2 The problem is concentrated

The ten categories with the most hand-offs produce 73% of them. **Category 23** stands out: 84.5% of its tickets change hands, and it produces 8.9% of all hand-offs from 4.6% of incidents.

3.3 Automated routing does not beat today's routing

Could a system, using only what is known when a ticket opens, send it to the group that eventually resolves it? Trained on earlier incidents and tested on 7,364 later ones (96% opened in one month), the answer is no. Today's first assignment reaches the resolving group **66.3%** of the time. A routing table built from history reaches 55.9%, machine learning 61.3%, and always picking the most common group 48.6%. On the incidents that really did bounce, today's routing was right only 12% of the time and machine learning 33%, but it gives that back on the many tickets humans route correctly.



Left: all categories, with 95% intervals. Right: Category 23 only, chosen on earlier data (section 3.4).

Robustness check	Router	Today	Incidents
Random 70/30 split (not how it would be deployed)	62.0%	66.1%	7,364
Main test split, with the caller and the agent who opened the ticket added as fields	61.7%	66.3%	7,364
Retrained on everything before 2016-04, tested on 2016-04	60.2%	68.9%	7,763
Retrained on everything before 2016-05, tested on 2016-05	62.8%	66.8%	7,474

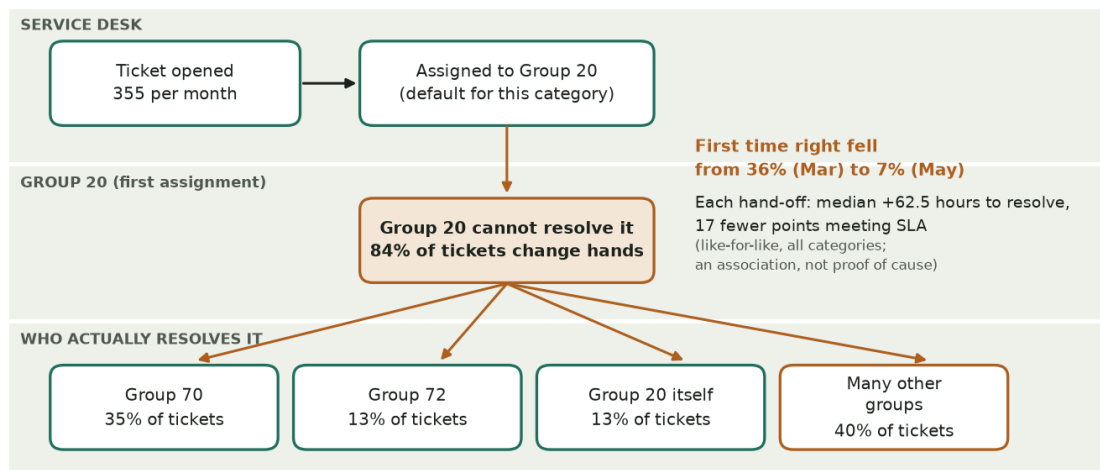
The result holds under every way of testing it, and adding the caller and the agent who opened the ticket did not help (61.7% vs 61.3% for the same model without them). The likely reason is that the fields recorded at

intake (100% of tickets have a category and 77% a symptom) do not contain what is needed to route many tickets. Diagnosis happens after the ticket is opened.

3.4 The exception: Category 23

Category 23 was the only category where, using earlier data alone, a rule clearly beat today's routing on a validation slice. Judged on the untouched later incidents, today's first assignment is right **6.7%** of the time and one rule (**Category 23 -> Group 70**) is right **42.4%** of the time, on 330 tickets: 118 more routed correctly than before. A simpler rule beat the routing table (35.2%).

Current state: Category 23 tickets, 2016-03 to 2016-05



Where Category 23 tickets start and where they end up.

Month	Tickets	Routed right today	Resolved by Group 70	Change hands	Gain from the rule
2016-03	350	35.7%	33.4%	71.7%	-2.3 pts
2016-04	359	18.1%	42.3%	84.1%	+24.2 pts
2016-05	356	6.7%	43.0%	96.1%	+36.2 pts

The problem is getting worse, and the data cannot say why. First-time-right fell from 36% to 7% in three months while the hand-off rate rose from 72% to 96%. The rule would have been slightly worse in March, so the benefit depends on the window. Something changed in how these tickets are routed; finding out what comes before any rule change.

3.5 A second lever to test, not yet to act on

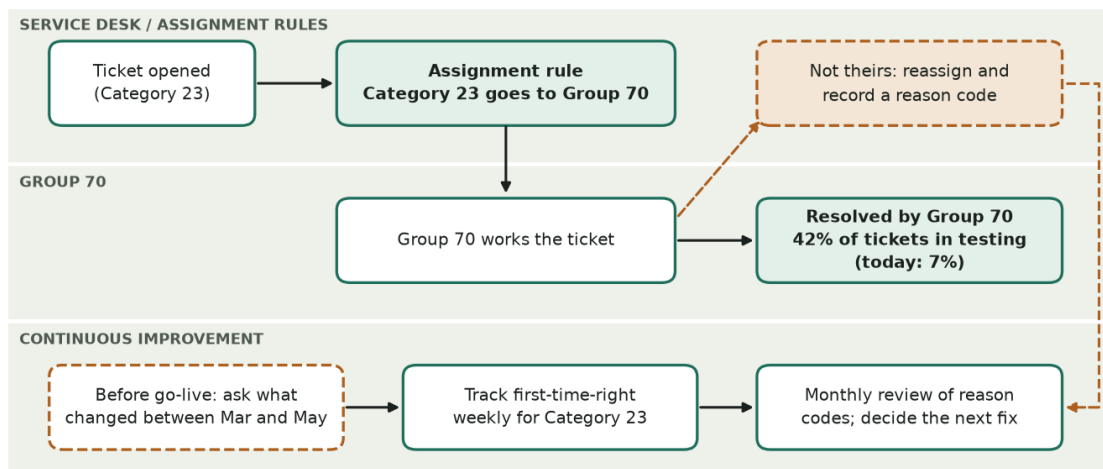
Hand-off rates vary between the 21 agents who open the most tickets, even after allowing for the categories they handle: from -11.6 to +16.8 points against expectation, and 13 of 21 differ by more than chance would explain. This points to differences in intake practice worth investigating, but ticket mix within a category may explain some of it, so it is a hypothesis for a pilot.

4. Options considered

Option	Evidence	Effort and risk	Verdict
A. Do nothing	41% of incidents change teams; Category 23 is getting worse (36% to 7% first time right)	No cost, but the cost continues	Not recommended
B. Machine-learning router, all categories	61.3% accuracy vs 66.3% today; worse under every test	Highest: build or buy, integrate, maintain (not costed here)	Reject
C. Routing rules for every category	55.9% vs 66.3% today	Moderate; would degrade routing that works	Reject
D. One rule for Category 23, reason codes, monthly review	42.4% vs 6.7% today on later incidents; chosen without seeing the test data	About 8 hours of configuration (assumption); risk that hand-offs move to Group 70	Recommend as a four-week pilot
E. Coach intake practice	13 of 21 agents differ beyond chance (association)	Low; confounding not ruled out	Investigate alongside D

5. Recommendation and what it changes

Proposed state: one assignment rule, then measure and review



The proposed path for Category 23 tickets, with the review loop.

The evidence for the change, separated by how far each piece can be trusted:

Evidence	Result	Basis
Category 23 routed to the right team, today vs the rule	6.7% vs 42.4%	Measured on later incidents
Incidents routed right instead of wrong, in that test window	118	Measured on later incidents
Whole-period gain (March to May), if the rule had always applied	+19.5 points	Measured, in-sample
Incidents newly routed right per month	69 to 127	Estimated from the gains above
Elapsed resolution hours saved per month	1,084 to 7,936	Assumption: 25% to 100% of the gap is caused by the hand-off
Analyst hours saved per month	12 to 80	Assumption: 10 to 20 minutes per hand-off
Analyst cost saved per year	\$4,853 to \$53,061	Assumption: \$35 to \$55 per hour, volume unchanged
Payback on the one-off change	3 to 21 days	Assumption: 8 hours of configuration

No fix has been deployed, so the shaded rows are arithmetic on stated assumptions, not results. The dollar savings are small; the case rests on elapsed time and SLA, which come from the data. The Excel model lets a stakeholder change every assumption.

6. Risks and limitations

- **The hand-off may only move.** Group 70 resolves 33% to 43% of Category 23 tickets month by month, so most of what it receives directly will still need passing on. The pilot tracks reason codes and Group 70's workload.
- **Group 20 may be a deliberate step** (for example a triage or security check) that the data cannot show.
- **The benefit depends on the window:** -2 points in March, +24 points in April, +36 points in May. The model's low case uses the whole-period figure.
- **Time and SLA gaps are associations.** The model applies a haircut of 25% to 100% to reflect that.
- **One log, one company, about three months.** Volumes and patterns may not hold elsewhere, and the routing fields are taken from the first record where each is filled in, which may be after some triage.
- **Labels are the group that finally resolved the ticket.** That is correct in hindsight, but not necessarily the best first assignment.

7. Next steps and decision rule

Four steps over about six weeks (detail in the requirements document): ask Group 20, Group 70 and the service desk what changed between March and May; build and test the rule and a rollback; go live and track first-time-right weekly; decide at week six. **Proposed rule, to be agreed with stakeholders:** keep the rule if Category 23 first-time-right is 30% or more, investigate at 15% to 30%, and roll back below 15% or if any other category's hand-off rate rises by 3 points or more.

Appendix: how the numbers were checked

- The hand-off counts were computed in pandas and recomputed independently in SQL with window functions; the two match exactly (19,088 hand-offs across all incidents, and the top ten categories).
- The Excel model was calculated by Excel itself and compared with the Python calculation: all 27 results match.
- Automated tests confirm no training incident is later than any test incident and that the router is never given a field that reveals the answer.
- Category selection used a validation slice of the training period only; the test set was not used to choose it.